INCH-POUND
MIL-DTL-16377/20B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/20A(SH)
18 March 1982

DETAIL SPECIFICATION SHEET

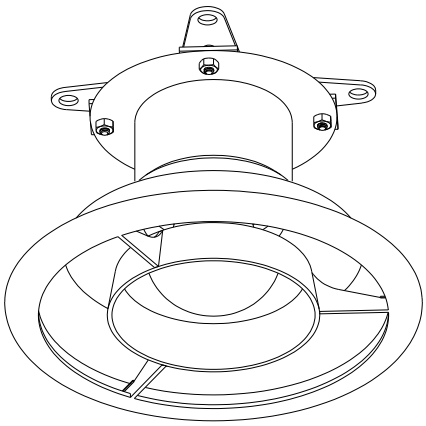
FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, 50 WATTS, 120 VOLTS, DRIPPROOF 15 DEGREES, SYMBOLS 64.1, 65, 64.1SSL, AND 65SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

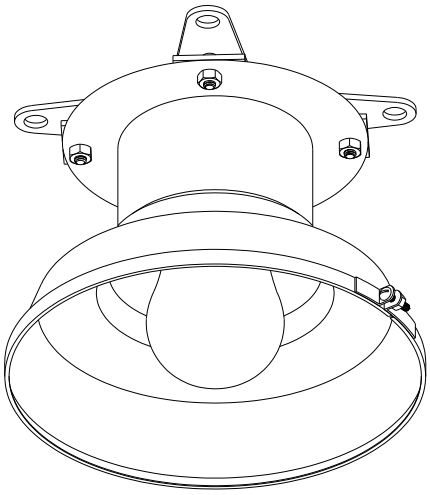
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 9 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.



SYMBOL 64.1



SYMBOL 65

FIGURE 1. Typical view of symbols 64.1 and 65.

AMSC N/A FSC 6210

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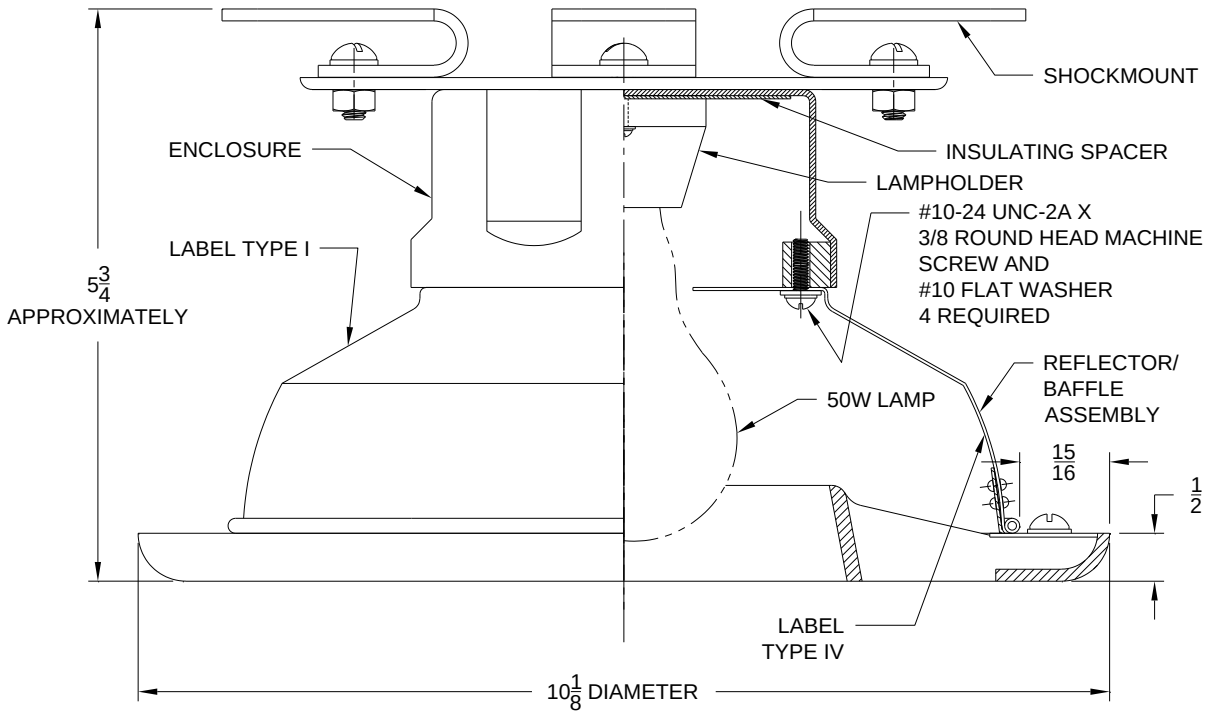


FIGURE 2. Fixture assembly, symbol 64.1.

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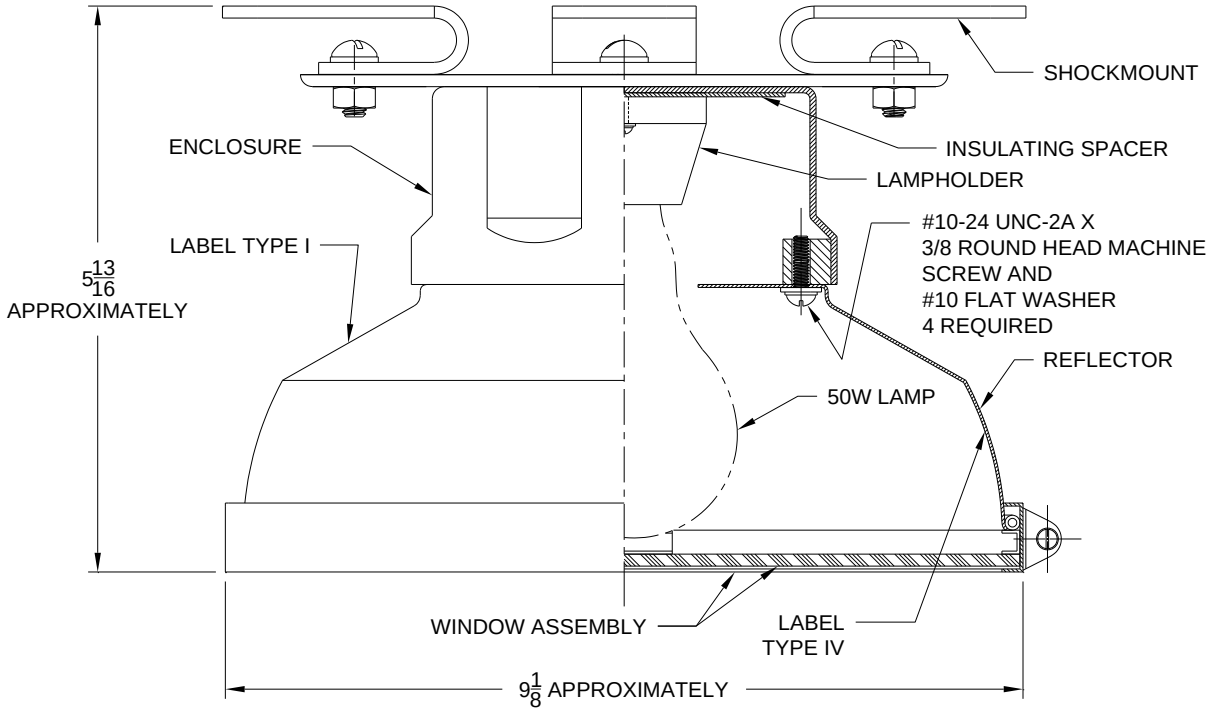


FIGURE 3. Fixture assembly, symbol 65.

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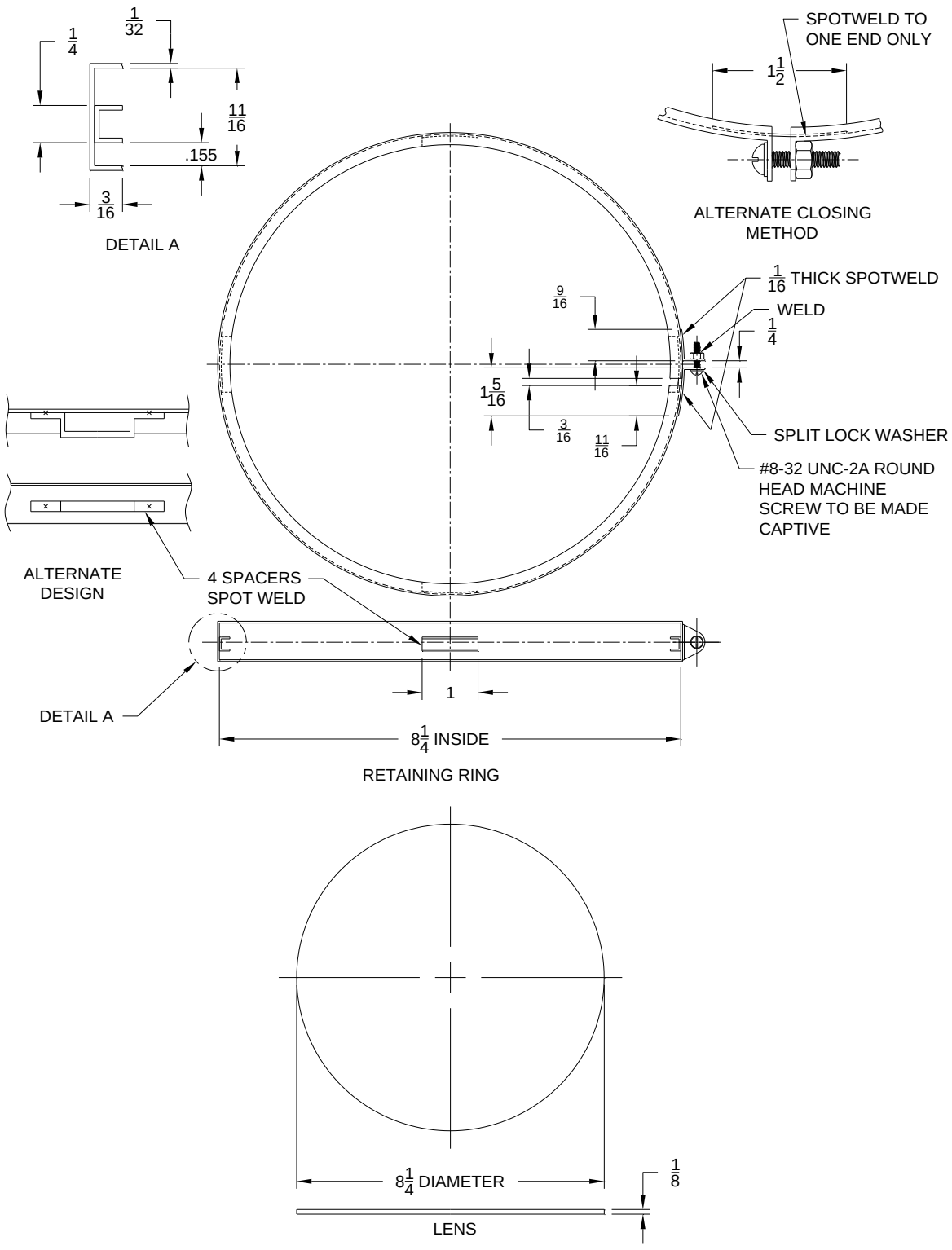


FIGURE 4. Window assembly.

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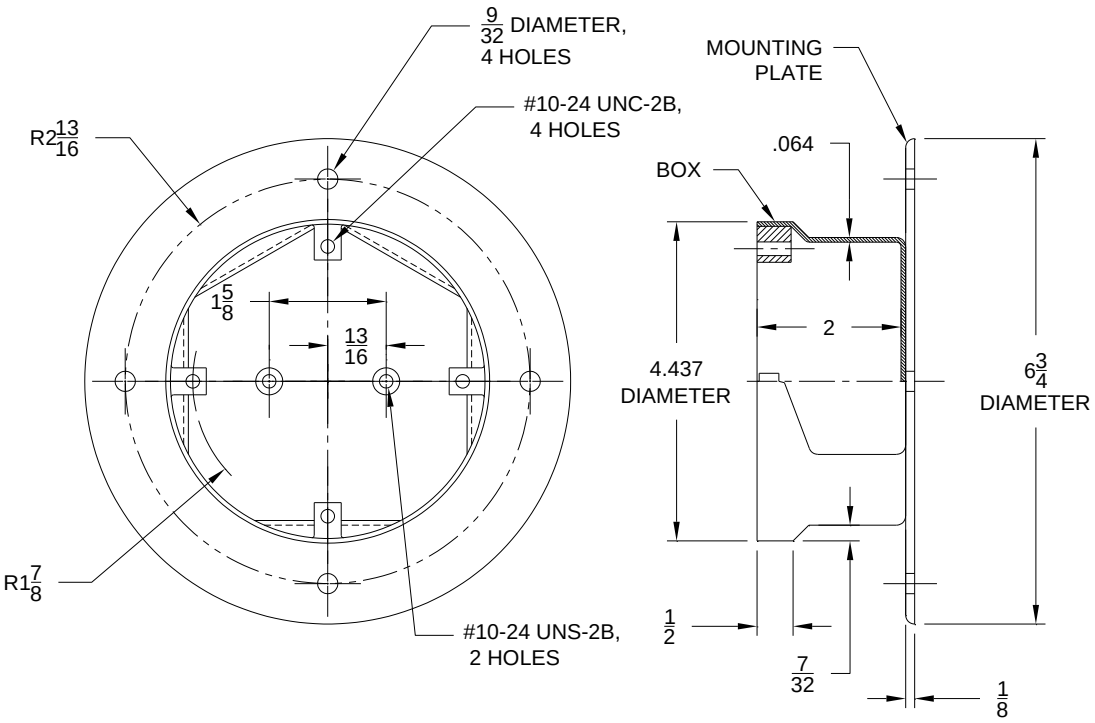


FIGURE 5. Enclosure.

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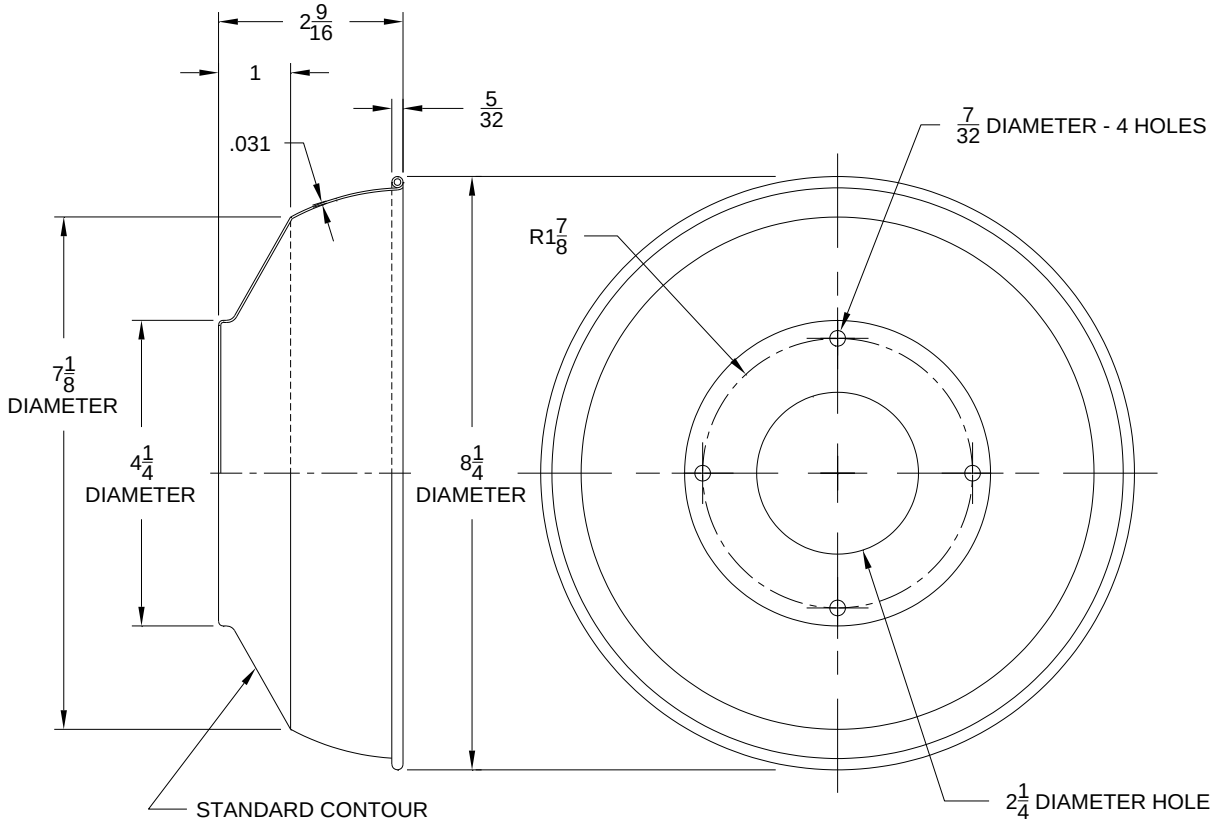


FIGURE 6. Reflector.

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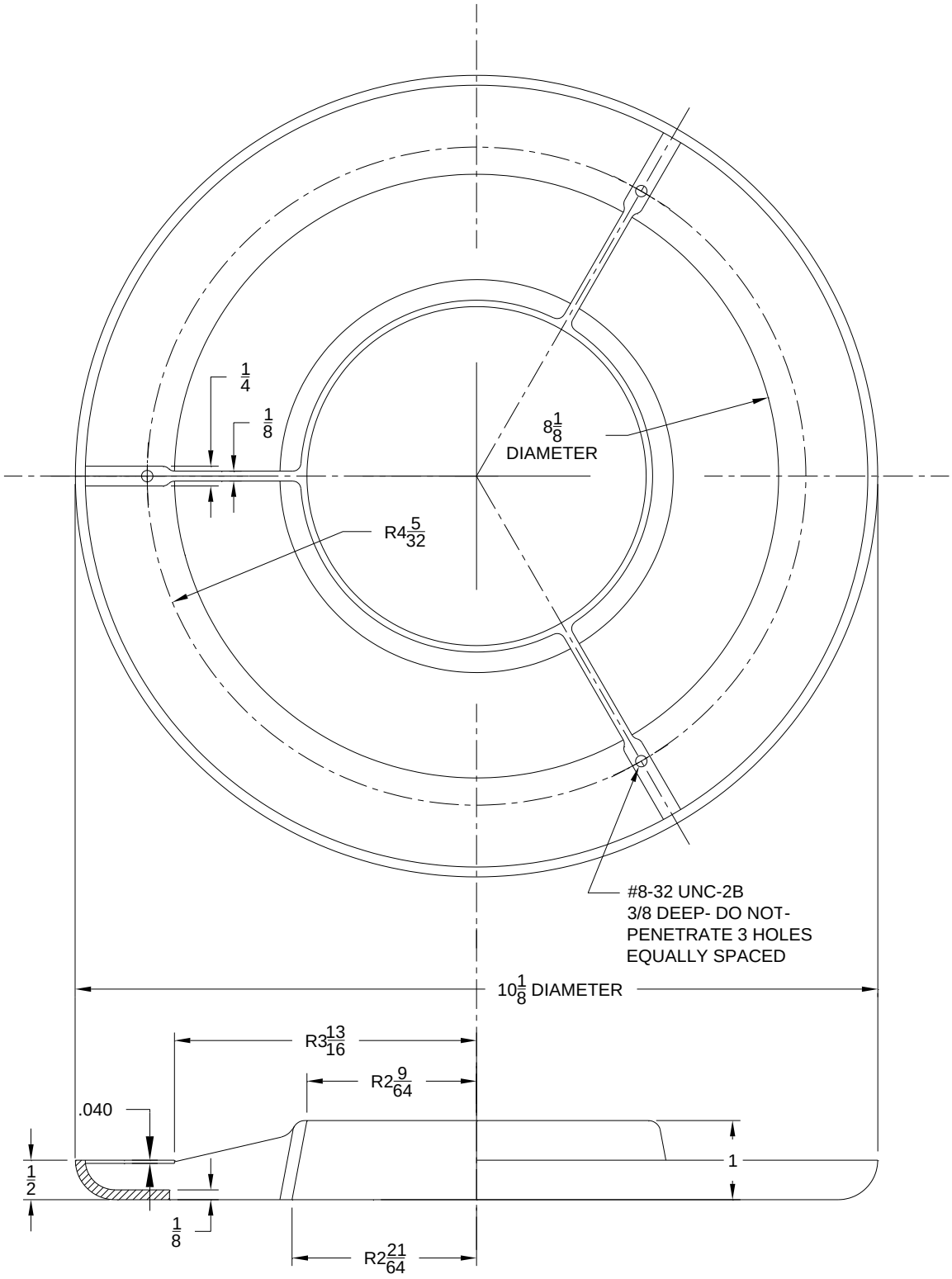


FIGURE 7. Baffle.

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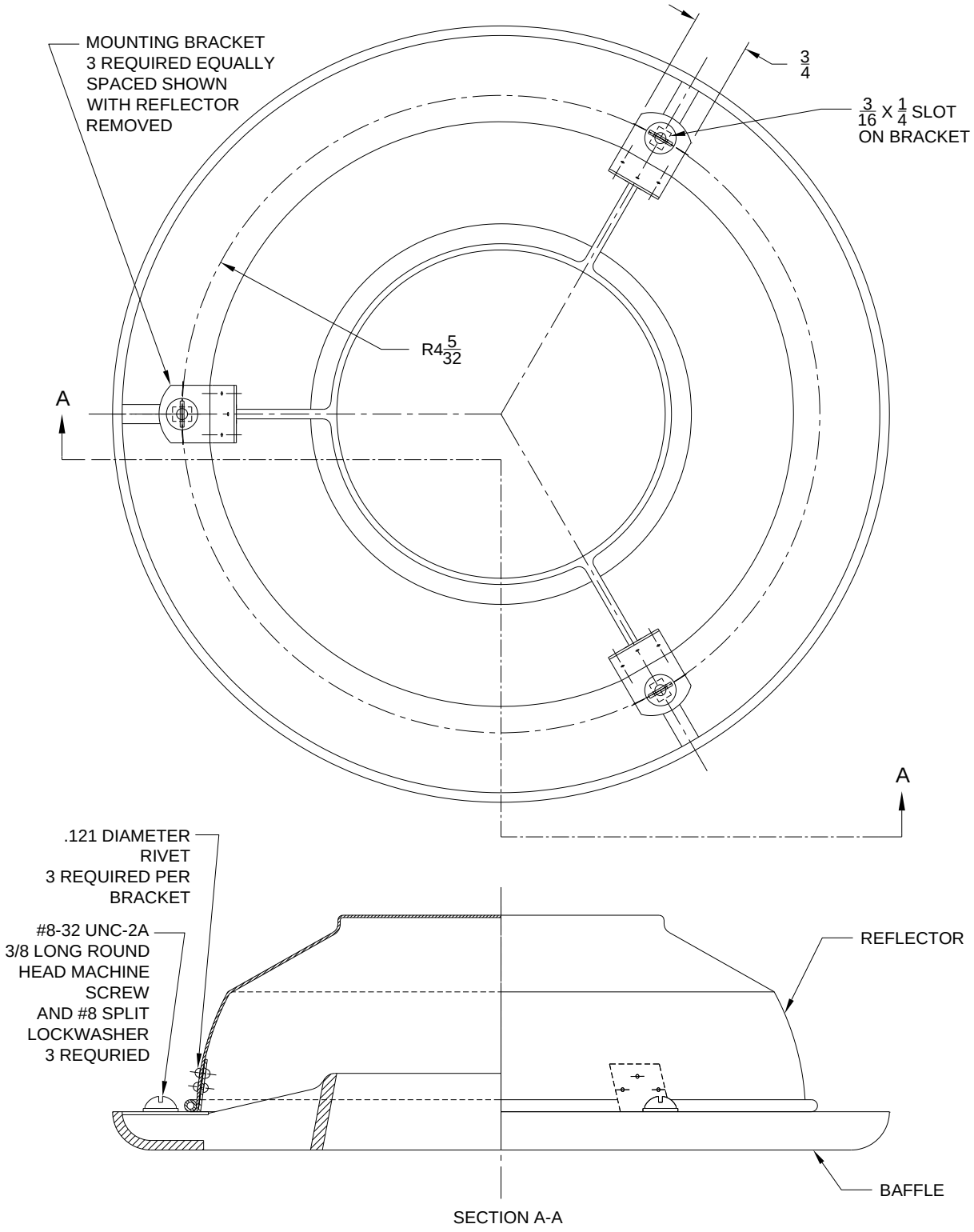
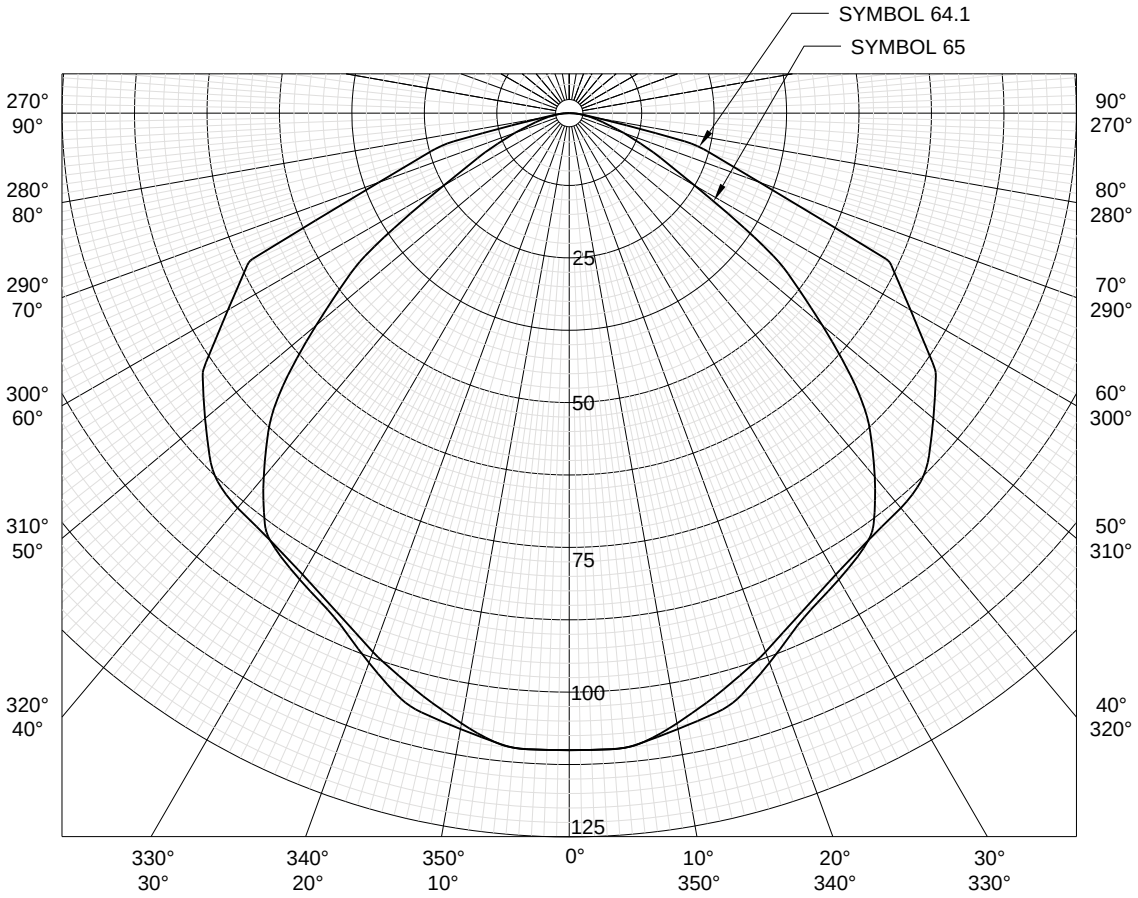


FIGURE 8. Reflector/baffle assembly.

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ANGLE	CANDLEPOWER	
	SYM 64.1	SYM 65
0	110	110
5	110	110
10	108	107
15	106	103
20	101	99
25	96	95
30	93	92
35	90	90
40	82	89
45	72	87

ANGLE	CANDLEPOWER	
	SYM 64.1	SYM 65
50	57	82
55	43	77
60	25	68
65	16	61
70	10	35
75	6	24
80	0	5
85	0	2
90	0	0
95	0	0

Photometric test data:

Distance: 10 feet

Voltage: 120 volts

Ambient temperature: 76 °F

Lamp: 50 watts, rough service conforming to A-A-50449, furnished by laboratory for test, shall be used.

FIGURE 9. Candlepower distribution curves.

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INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 2

SYMBOL NUMBERS

64.1: Complete lighting fixture assembly with reflector and baffle assembly.
65: Complete lighting fixture assembly with reflector and window assembly.

DESIGN

Electrical rating:	50 watts, 120 volts.
Dimensions and configurations:	See figures 1 through 8.
Mounting dimensions:	See figure 28 of MIL-DTL-16377.
Photometric data:	See figure 9.
Enclosure assembly:	Material for enclosure and reflector shall be aluminum.
Window assembly (symbol 65 only):	See figure 4. Material for lens shall be polycarbonate, clear, with light transmittance exceeding 80 percent, and shall conform to the property values of ASTM D3935. A retaining ring as shown on figure 4 shall be provided for attaching the window to the reflector rim. All metal parts of the retaining ring shall be steel, zinc plated.
Lampholder:	UL 496, Supplement SB, single-contact, medium-screw.
Baffle assembly (symbol 64.1 only):	Aluminum, cast or fabricated.
Shock mounts:	MIL-F-16377/41, part number M16377/41-003.
Enclosure:	Dripproof.
Finish:	All surfaces, except shock mounts and window assembly, shall receive a baked white enamel finish.
Weight:	
Symbol 64.1:	2 pounds, 4 ounces maximum.
Symbol 65:	2 pounds, 0 ounces maximum.
Lamp:	These fixtures shall be designed to accommodate a 50-watt, 120-volt, A-19 bulb, inside frosted, medium-screw base, rough service lamp conforming to W-L-101/80. (Lamp not furnished.)
Designation and markings:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 2 and 3: (a) One type I. (b) One type IV. Apply one type IV label (as shown on figures 2 and 3), with the following information in 3/16-inch letters: "USE LAMP NSN 6240-00-143-3070 ONLY."

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBERS

64.1SSL: Complete SSL fixture assembly.
65SSL: Complete SSL fixture assembly.

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DESIGN

Electrical rating:	25 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 and 3.
Photometric data:	See figure 9 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Mounting dimensions (symbol 64.1SSL):	See figure 2 and figure 28 of MIL-DTL-16377.
Enclosure:	Dripproof. See 3.5.11 of MIL-DTL-16377.
Weight:	
Symbol 64.1SSL:	2 pounds, 4 ounces maximum.
Symbol 65SSL:	2 pounds, 0 ounces maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 2 and 3.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I for type II and table III for type III.
Repair parts:	See table II.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table IV.

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TABLE I. First article and comparison inspection for complete type II fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Shock	3.5.12	4.8	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof
Continuity of grounding	3.6.13.4	4.8.12	---
Optical uniformity	3.7.4.11	4.8.4	Symbol 65 only
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	---

TABLE II. First article and comparison inspection for repair parts.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	1/
Light output	3.7.3.7	4.8.6.3	Luminous transmittance 1/

NOTE:

1/ For lens of window assembly only.

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TABLE III. First article and comparison inspection for complete type III fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14.4	1/ Dripproof, unless otherwise specified.
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14.4	1/ Dripproof, unless otherwise specified.
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	1/
Photometrics	3.7.1	4.8.6	---
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. Part numbers and general information.

Military Part Number M16377/20-	Item Name	National Stock Number (NSN)	Remarks
64.1	Complete fixture, symbol 64.1	6210-00-295-2881	Figure 2
65	Complete fixture, symbol 65	6210-00-505-0115	Figure 3
64.1SSL	Complete fixture, solid state, symbol 64.1SSL	---	---
65SSL	Complete fixture, solid state, symbol 65SSL	---	---
001	Window assembly	6210-00-897-6976	Figure 4
002	Reflector	6210-00-295-2681	Figure 6
003	Reflector/baffle assembly	6210-00-399-6163	Figure 8

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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